

# Abstract Algebra - Homework 2

Simon Gustafsson

## Problem 1

(1)

$$ij = k, \quad ij = -ji \implies -ji = k \implies ji = -k$$

Hence,  $ij \neq ji$ , so multiplication is not commutative and  $\mathbb{H}$  is a non-commutative ring.

- (2) **(a)** Let  $x, y \in \mathbb{H}$ . One can verify by direct computation that conjugation reverses the order of multiplication,  $\overline{xy} = \bar{y}\bar{x}$ . Note that for any  $h = a + bi + cj + dk \in \mathbb{H}$ , we have

$$N(h) = h\bar{h} = a^2 + b^2 + c^2 + d^2 \in \mathbb{R},$$

so  $N(h)$  lies in the center of  $\mathbb{H}$  and commutes with all elements. Using this, we compute:

$$\begin{aligned} N(xy) &= (xy)\overline{(xy)} \\ &= (xy)(\bar{y}\bar{x}) \\ &= x(y\bar{y})\bar{x} \\ &= xN(y)\bar{x} \\ &= N(y)x\bar{x} \\ &= N(y)N(x) = N(x)N(y). \end{aligned}$$

- (b)** Since  $N(x) = x\bar{x} = a^2 + b^2 + c^2 + d^2$ , it follows that

$$N(x) = 0 \iff a = b = c = d = 0 \iff x = 0.$$

Thus,  $N(x) = 0$  if and only if  $x = 0$ .

- (3) Let  $x \in \mathbb{H}$  be nonzero. From previous section, we have  $N(x) = x\bar{x} = a^2 + b^2 + c^2 + d^2 > 0$ . Define the inverse of  $x$  by

$$x^{-1} := \frac{\bar{x}}{N(x)}.$$

Then:

$$xx^{-1} = x \cdot \frac{\bar{x}}{N(x)} = \frac{x\bar{x}}{N(x)} = \frac{N(x)}{N(x)} = 1, \quad x^{-1}x = \frac{\bar{x}}{N(x)} \cdot x = \frac{\bar{x}x}{N(x)} = \frac{N(x)}{N(x)} = 1.$$

Hence, every nonzero element of  $\mathbb{H}$  is invertible.

Suppose  $x, y \in \mathbb{H}$  with  $x \neq 0$  and  $xy = 0$ . Then,  $x^{-1}(xy) = (x^{-1}x)y = 1 \cdot y = y = 0$ . Thus,  $y = 0$ , and similarly one shows that  $x = 0$  if  $y \neq 0$  and  $xy = 0$ . Therefore,  $\mathbb{H}$  has no zero-divisors, and every nonzero element is invertible.